

Operating Systems and Kernel Design
EECE 4811/EECE 5881
Fall 2023
Prof. Megherbi
Assignment #1

In this assignment, you will be working with processes on the Linux platform. You will be programming in C or C++

Part1. a) Write a program ***prog*** that repeatedly creates a sub-process using `fork`, and waits until it terminates. Each child process displays “Child Running” and calls `exit` immediately.

b) modify ***prog*** so that each child process spawns another program *b* using `execve`. The program *b* displays “I am the grandchild and immediately exits.

Part2. Write a program that takes a single integer argument *n* from the command line and creates a set of $2*n$ processes ($n \leq 4$). In a loop for $k=1:2*n$, each process should display the phrase “ I am process *x* of process *k*”, where *x* is the process ID and *k* is the process number *k*, and then the program terminates.

For example, if the user enters the command:

% run 4 (Where run is the executable of your c program.)

Then, the output of your program should be:

```
I am process_ID of process 1
I am process_ID of process 2
I am process-ID of process 3
I am process_ID of process 4
.
.
.
I am process_ID of process 8
```

Note: Ensure that the original parent process does not terminate until all of its children have died (make careful use of the `wait()` function as seen in class). As seen in class, if you create zombies, you need to kill them using the appropriate system command as discussed in class).